

Correlation Analysis of Life Expectancy and Different Variables

DTSC 610 M01

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Jungmin Sung

Project Goal & About Dataset

Goal:

- Identify any variables with significant correlation coefficient vs Life Expectancy in this project. ± 0.70

Dataset:

- World Health Statistics 2020 Report (Filtered/Cleaned version on Kaggle.com)
 - 39 categories covering the health related statistics - 39 separate files.
(each column average: 184 countries, periods of 2000-2019, male/female/both stats provided)
- Link:
 - <https://www.who.int/data/gho/publications/world-health-statistics>
 - <https://www.kaggle.com/utkarshxy/who-worldhealth-statistics-2020-complete/metadata>



Correlation

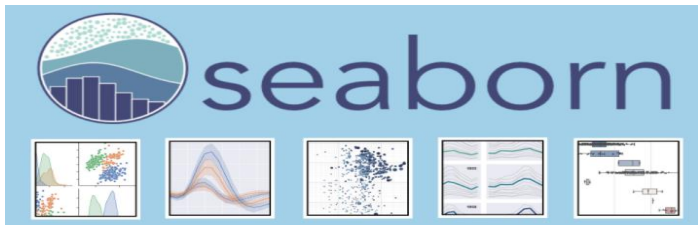
- The correlation is **not** the evidence of cause-effect relationship. However, it can be a good starting point to investigate on the existence of such relationship.
⇒ Then, identifying How and Why.
- The identified correlations could be used for the future studies on the real life relationship between Life Expectancy and these variables.

library/package

import glob $\Rightarrow$ for reading CSV files.

import pandas **as** pd $\Rightarrow$ for manipulating data frames & `.corr()`

import seaborn **as** sns $\Rightarrow$ for visualization



Dataset preprocessing

- Life expectancy == Average living years == Average age of death
 - 184 countries, years: 2000, 2010, 2015, 2019, Age of male/female/both.
- Dataset:
 - Good:
 - Wide range of years (mostly 2000-2019), stats for each country, separate stats for male/female/both.
 - Bad:
 - The time periods and the number of country are not always matched with another categories,
 - Merging on Country & Year will have much less data samples.

Dataset preprocessing

- After filtered/cleaned:
 - 23 tables with 2 columns: Country / Stat of both in year 2015
 - Then, categories in similar fields are merged together → 9 tables.

# of Countries						
Group # 1	LifeExp	Malaria	TB	HepB	NTD	HIV
Group # 2	LifeExp	doctors	nursing_mid	dentists	Pharmacists	
Group # 3	LifeExp	drinking_water	sanitation	safe_san	handwash	
Group # 4	LifeExp	MR_infant_28d	MR_infant_1y	MR_infant_5y		
Group # 5	LifeExp	cardio_cancer	suicideRate			
Group # 6	LifeExp	alcohol	Tobacco			
Group # 7	LifeExp	UHC_cov				
Group # 8	LifeExp	MR_Poison				
Group # 9	LifeExp	clean_fuel_tech				

	LifeExp	Malaria	TB	HepB	NTD	HIV
0	61.65	14.91	189.0	0.50	13765392	0.03
1	76.54	0.00	75.0	0.22	7561	0.05
2	61.72	200.00	366.0	4.85	18534219	0.95
3	76.17	0.00	26.0	0.01	5406	0.14
4	74.51	0.00	50.0	0.25	39044	0.06
...	...	...	...	...	...	...
81	74.66	10.57	30.0	0.62	280243	0.20
82	73.16	0.17	199.0	1.20	4468764	0.11
83	67.47	24.90	48.0	2.54	5840695	0.04
84	60.50	202.50	391.0	1.84	11830416	4.11
85	58.48	97.65	242.0	4.38	10346543	3.91

Correlation in Communicable Diseases Group

- This group is in negative correlation with life expectancy.
 - Countries with less communicable diseases incidents per year have higher # in Life Expectancy.
 - Countries with higher # in life expectancy have less c.d. Incidents per year.

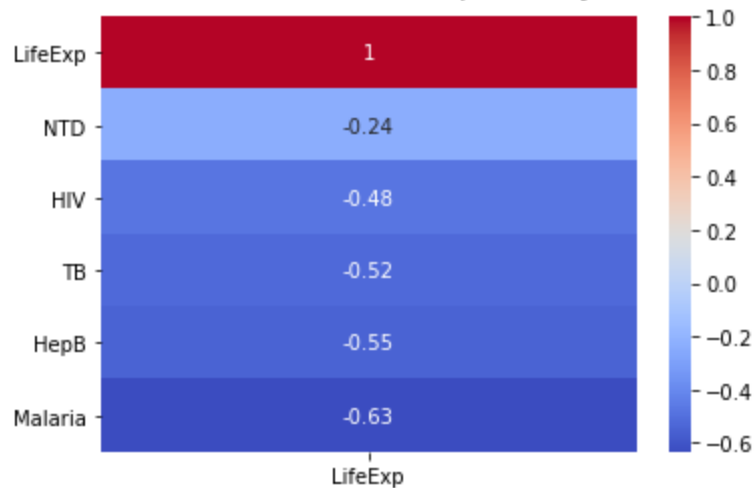
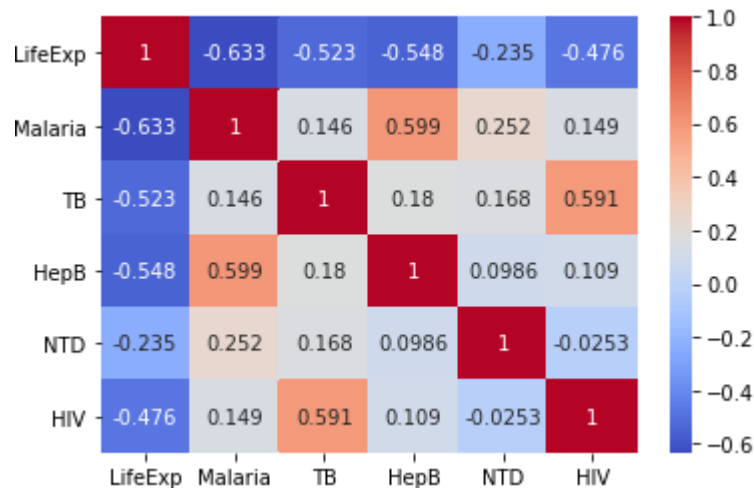
	LifeExp	Malaria	TB	HepB	NTD	HIV
LifeExp	1.000000	-0.633123	-0.523186	-0.547870	-0.235096	-0.476241
Malaria	-0.633123	1.000000	0.146096	0.598863	0.251594	0.148704
TB	-0.523186	0.146096	1.000000	0.179987	0.167929	0.591151
HepB	-0.547870	0.598863	0.179987	1.000000	0.098615	0.109069
NTD	-0.235096	0.251594	0.167929	0.098615	1.000000	-0.025340
HIV	-0.476241	0.148704	0.591151	0.109069	-0.025340	1.000000

Visualization of Correlation in Heatmap

- Seaborn Heatmaps:

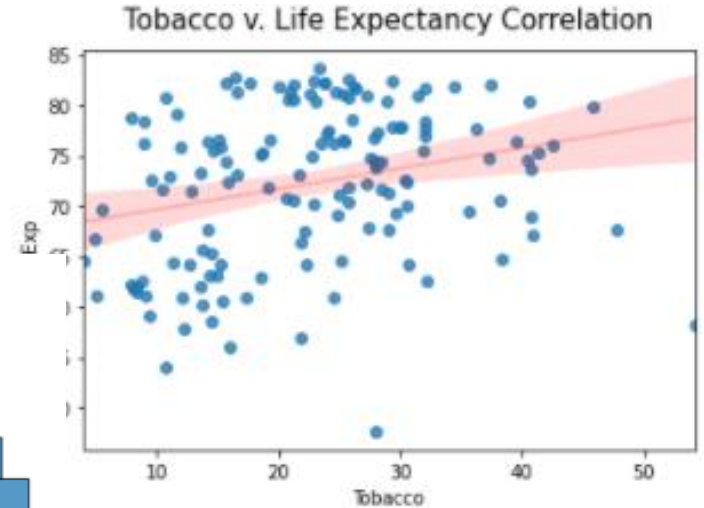
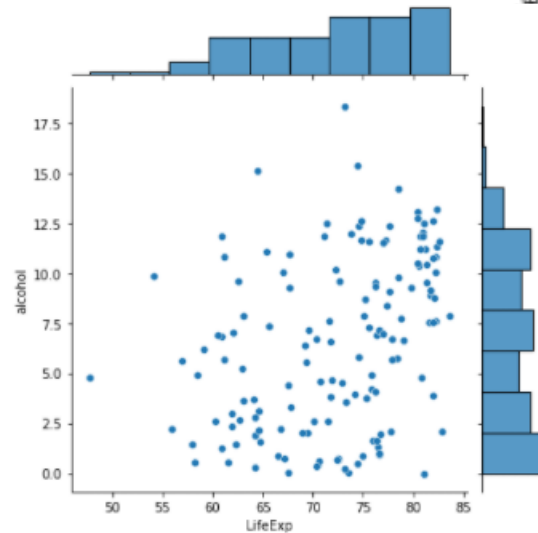
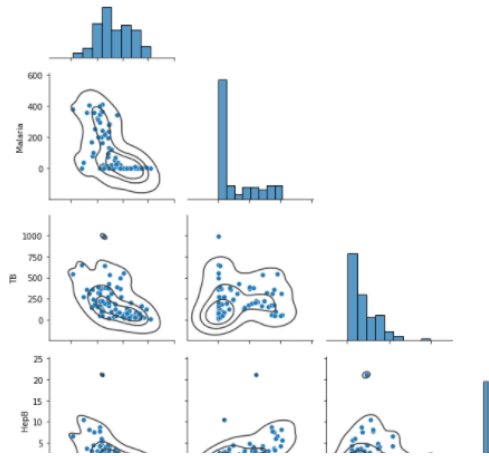
```
sns.heatmap(dataframe.corr(), annot=True, fmt='.3g', cmap='coolwarm')
```

```
sns.heatmap(dataframe.corr()[['col']].sort_values(by='col') ...
```



Visualization of Correlation in Plots

- Seaborn Pairplot / Jointplot / Regplot ... etc



Correlation: Life Expectancy vs One-by-One

- Instead of finding correlations in group, finds correlation between life expectancy samples and (ex) Malaria samples before merged into groups. $\Rightarrow$ More samples, Different Corr.

Variable	Correlation	Interp.	+/-
21 MR_infant_1y	-0.895958	Strong	Negative
22 MR_infant_5y	-0.874980	Strong	Negative
20 MR_infant_28d	-0.871679	Strong	Negative
19 UHC_Cov $\rightarrow$ k	0.865219	Strong	Positive
11 sanitation	0.822244	Strong	Positive
12 drinking_water	0.814540	Strong	Positive
10 clean_fuel_tech	0.793851	Strong	Positive
9 MR_Poison	-0.787836	Strong	Negative
13 handwash	0.782170	Strong	Positive
7 alcohol	0.373452	Weak	Positive
8 Tobacco	0.265485	Weak	Positive
17 nursing_mid	0.182041	Weak	Positive
6 suicideRate	0.179331	Weak	Positive
3 NTD	-0.111964	Weak	Negative

14	safe_san	0.694502	Moderate	Positive
18	doctors	0.690216	Moderate	Positive
15	dentists	0.684040	Moderate	Positive
16	Pharmacists	0.681462	Moderate	Positive
1	TB	-0.670218	Moderate	Negative
5	cardio_cancer	-0.662305	Moderate	Negative
23	birth_att_skilled	0.661455	Moderate	Positive
0	Malaria	-0.650912	Moderate	Negative
2	HepB	-0.569489	Moderate	Negative
4	HIV	-0.534991	Moderate	Negative

86 samples	LifeExp	1.000000
	Malaria	-0.633123
	TB	-0.523186
	HepB	-0.547870
	NTD	-0.235096
	HIV	-0.476241

183 samples $\leftarrow$ (pointing to TB in the middle table)

107 samples $\leftarrow$ (pointing to Malaria in the middle table)

183 samples $\leftarrow$ (pointing to HepB in the middle table)

118 samples $\leftarrow$ (pointing to HIV in the middle table)

183 samples $\leftarrow$ (pointing to NTD in the first table)

Strong Correlations with Life Expectancy

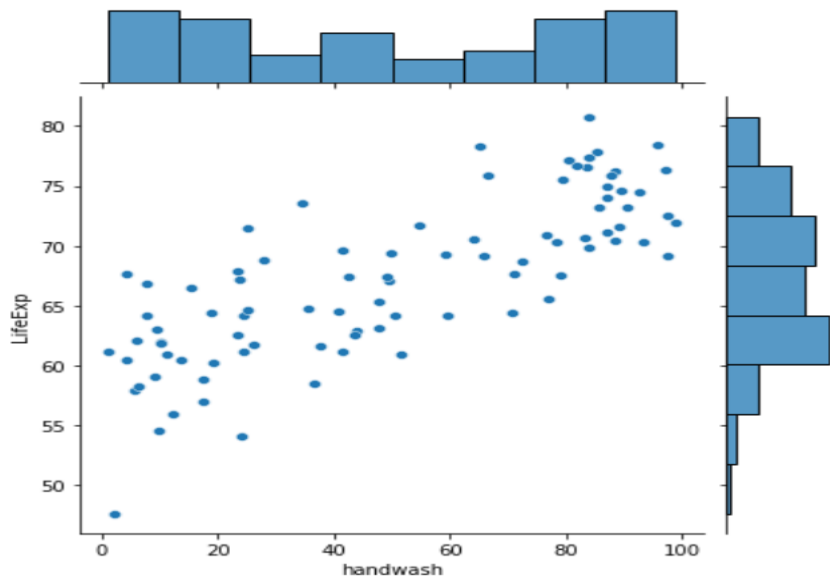
- Basic Hand washing Facilities at Home + + +
- Drinking Water Services + + +
- Universal Health Coverage (UHC) + + +
- Population with Primary Reliance on Clean Fuels and Technologies + + +
- Mortality Rates Newborn and Child - - -
- Mortality Rates of Unintentional Poisoning - - -

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Possible Future Studies

- Basic Hand washing Facilities at Home + + +

Population with basic handwashing facilities at home (%) v. Life Expectancy



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